HOYA

HOYA OPTICAL GLASS

HOYA

HOYA GLASS WORKS, LTD.

572 Miyazawa-cho, Akishima-shi, Tokyo, JAPAN

Tel : (0425) 45-1031

Cable : HOYAGLAS TOKYO

Telex : J26585 HOYAGW

HOYA

HOYA OPTICAL GLASS

Catalog No. 7509E

Hoya Glass Works, Ltd. was established in 1941 as a manufacturer of optical glass and, later, crystal glass and spectacle lens. It has won a reputation for the excellent quality of these products, and thus enjoys a large and growing share in each of the product markets.

The Optical Glass Division is continually developing new techniques and products, through research and development in the Hoya Technical Research Center. In 1965, Hoya succeeded in developing a method for the continuous melting of optical glass which assures a high quality product in this era of technical innovation. Hoya's contribution to the optical industry extends also to scientific, industrial, and electronic applications, by offering glasses to meet specific needs of today's demanding market, to include various types of colored glasses and electro-optical glasses.

This catalog is revised in accordance with the newly revised specifications of the Japanese Optical Industrial Standards to include various recently developed abnormal-dispersion glasses and cadmium-, thorium-free glasses. Please note the marked improvement in abrasion performance, chemical durability, and coloration in our current glasses.

September 1975
Hoya Glass Works, Ltd.

MAIN PRODUCTS
(Optical Division)

- Optical Glass
- Colored Filter Glass
  - Photographic Filter
  - Glass for Scientific and Technical Uses
- Lens and Prism Blank for Binoculars
- Spectacle Lens
- Glass for Industrial Uses
  - Large Size High Precision Optical Glass
  - Low Expansion Crystallized Glass
  - Gage Glass
  - Chemically Strengthened Glass
  - Watch Crystal Glass
- Glass for Electronic Uses
  - Ultrasonic Delay Line Glass
  - Acousto-optic Tellurite Glass
  - High Power Laser Glass
  - Glass Substrate for I.C. Photomasks
  - Others

CONTENTS

1. DESIGNATION OF GLASS TYPE

Optical glasses are classified by their main chemical components and are identified by refractive index (n_d) and Abbe-number (ν_d). They are divided into groups. Each glass type within a group is designated by the abbreviated group symbol and a number. For example, boro-silicate crown 7 glass is designated as BSC 7 (BK 7 in the customary designation). In addition to our glass type designation, a six-digit code number is listed in this catalog. The first three digits indicate the n_d after the decimal point, and the last three digits represent the ν_d. In BSC 7, for example, the n_d is 1.51680, the ν_d is 64.20, for which we indicate as 517-642. The code numbers listed on pages 8 and 9 are arranged in order of increasing index value.

Illustrated on the last page is a diagram of n_d versus ν_d for all glasses listed in this catalog.

Group Designation

Group	Our Symbol	Customary Symbol	Group	Our Symbol	Customary Symbol
Fluor Crown	FC	FK	Extra Light Flint	FEL	LLF
Phosphate Crown	PC	PK	Barium Flint	BaF	BaF
Special Phosphate Crown	PCS	PKS	Light Flint	FL	LF
Dense Phosphate Crown	PCD	PSK	Flint	F	F
Boro Silicate Crown	BSC	BK	Dense Barium Flint	BaFD	BaSF
Light Barium Crown	BaCL	BaLK	Dense Flint	FD	SF
Crown	C	K	Special Dense Flint	FDS	SFS
Zinc Crown	ZnC	ZK	Fluor Flint	FF	TiF
Barium Crown	BaC	BaK	Light Lanthanum Flint	LaFL	(LaLF)
Dense Barium Crown	BaCD	SK	Lanthanum Flint	LaF	LaF
Extra Dense Barium Crown	BaCED	SSK	Niobium Flint	NbF	(NbF)
Light Lanthanum Crown	LaCL	(LaLK)	Tantalum Flint	TaF	(TaF)
Lanthanum Crown	LaC	LaK	Dense Niobium Flint	NbFD	(NbSF)
Tantalum Crown	TaC	(TaK)	Dense Tantalum Flint	TaFD	(TaSF)
Crown Flint	CF	KF	Abnormal Dispersion Crown	ADC	—
Antimony Flint	SbF	KzF	Abnormal Dispersion Flint	ADF	—
Light Barium Flint	BaFL	BaLF			

Code／Glass Type

Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type
465-658	FC 3	521-528	SbF 5	541-472	FEL 2	574-564	BaC 6	603-424	BaFD 5	615-512	BaCED 3
471-673	FC 1	522-595	C 5	548-458	FEL 1	575-415	FL 7	603-607	BaCD 14	617-366	F 4
487-704	FC 5	523-510	CF 5	549-454	FEL 7	578-417	FL 4	606-379	F 15	617-539	BaCED 1
498-651	BSC 3	523-586	C 12	551-495	SbF 1	580-537	BaFL 4	606-439	BaF 4	618-552	BaCED 4
501-563	C 10	525-646	PC 3	552-634	PCD 3	581-409	FL 5	607-404	BaFD 3	620-363	F 2
504-669	PC 1	526-511	CF 2	557-586	BaC 5	582-420	FL 3	607-493	BaF 5	620-381	F 9
508-613	ZnC 7	526-601	BaCL 1	560-471	FEL 3	583-465	BaF 3	607-567	BaCD 2	620-498	BaCED 9
510-634	BSC 1	527-511	SbF 6	561-452	FEL 4	583-595	BaCD 12	607-595	BaCD 7	620-603	BaCD 16
511-510	FF 1	529-516	SbF 2	564-608	BaCD 11	589-410	FL 2	608-462	BaF 7	620-622	ADC 1
511-605	C 7	531-621	BSC 6	565-530	ADF 1	589-485	BaF 6	609-589	BaCD 3	621-360	F 11
515-546	CF 3	532-488	FEL 6	567-428	FL 6	589-529	BaFL 6	610-567	BaCD 1	622-531	BaCED 2
517-522	CF 6	533-459	FF 2	569-560	BaC 4	589-613	BaCD 5	611-558	BaCD 8	623-569	BaCD 10
517-642	BSC 7	533-581	ZnC 1	569-631	PCD 2	592-583	BaCD 13	613-370	F 3	623-581	BaCD 15
517-697	PCS 1	534-515	CF 4	571-509	BaFL 2	593-355	FF 5	613-449	ADF 4	624-471	BaF 8
518-590	C 3	534-554	ZnC 5	571-530	BaFL 3	596-392	F 8	613-586	BaCD 4	625-356	F 7
518-604	BaCL 3	540-510	CF 1	573-426	FL 1	597-553	ADC 2	614-551	BaCD 9	626-357	F 1
518-652	PC 2	540-597	BaC 2	573-575	BaC 1	603-380	F 5	614-564	BaCD 6	626-391	BaFD 1

Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type	Code	Glass Type
636-353	F 6	658-573	LaC 11	686-494	LaFL 1	720-293	FD 20	755-275	FD 4	802-443	NbFD 14
639-451	BaF 12	664-359	BaFD 2	687-429	ADF 8	720-420	LaFL 6	757-318	NbFD 9	804-465	TaF 3
639-555	BaCD 18	664-489	BaF 21	689-312	FD 8	720-439	LaFL 5	757-363	NbFD 6	805-255	FD 6
640-346	FD 7	665-534	LaCL 3	691-547	LaC 9	720-503	LaC 10	757-477	NbF 2	805-396	NbFD 3
640-602	LaCL 6	667-331	FD 19	694-508	LaCL 5	722-292	FD 18	762-266	FD 14	806-407	NbFD 13
641-568	LaCL 1	667-483	BaF 11	694-533	LaC 13	723-380	BaFD 8	762-269	FDS 5	807-355	NbFD 8
643-479	BaF 9	668-419	BaFD 6	697-485	LaFL 2	726-534	TaC 1	762-403	NbFD 5	815-442	TaFD 1
643-580	LaC 6	669-449	BaF 13	697-555	LaC 14	728-283	FD 10	773-496	TaF 1	816-445	TaFD 10
648-338	FD 2	670-472	BaF 10	697-561	LaC 15	728-287	FDS 2	783-361	NbFD 7	831-365	NbFD 4
648-338	FD 12	670-517	LaCL 4	699-301	FD 15	734-511	TaC 4	785-257	FD 11	834-373	NbFD 10
649-530	BaCED 20	670-573	LaCL 7	700-349	BaFD 14	735-498	NbF 3	785-261	FDS 3	835-430	TaFD 5
650-339	BaFD 10	673-322	FD 5	700-478	LaFL 3	740-282	FD 3	786-439	NbFD 11	838-420	TaFD 2
650-557	LaCL 2	673-506	LaCL 9	702-412	BaFD 7	741-278	FD 13	788-475	TaF 4	847-238	FDS 9
651-383	BaFD 4	678-534	LaCL 8	713-433	LaFL 4	741-526	TaC 2	789-428	NbFD 1	850-322	TaFD 9
652-584	LaC 7	678-555	LaC 12	713-539	LaC 8	743-492	NbF 1	795-454	TaF 2	855-366	TaFD 13
654-337	FD 9	683-447	BaF 22	717-295	FD 1	744-449	LaF 2	797-411	NbFD 2		
658-509	BaCED 5	686-439	BaF 20	717-480	LaF 3	750-350	LaF 7	800-423	NbFD 12		

2. OPTICAL PROPERTIES

2.1 Refractive Index

Refractive indices to 5 decimals are given for the following 10 lines of the spectrum.

Spectral Line	t	r	C	D	d	e	F	g	h	i
Element	Hg	He	H	Na	He	Hg	H	Hg	Hg	Hg
Wavelength in nm	1,013.98	706.52	656.27	589.29	587.56	546.07	486.13	435.84	404.66	365.01

Each refractive index represents the mean of several melts measured at room temperature.

2.2 Dispersion

The main dispersion is expressed by $n_F - n_C$. The Abbe-number is defined:

$$\nu_d = \frac{n_d - 1}{n_F - n_C}$$

Also listed is the ν_e value:

$$\nu_e = \frac{n_e - 1}{n_{F'} - n_{C'}}$$

3. CHEMICAL PROPERTIES

In the processes of polishing, coating, and storing, optical glass is usually exposed to liquid water, to high humidity atmosphere, to temperature variation or to a combination of the three.

Polished glass exposed to high humidity as well as to temperature variation may "sweat". Water vapour may condense to form droplets on the glass surface. Some of the glass components that dissolve in the droplets, in turn may attack the glass surface and react with gaseous elements in the air (CO_2 for example). Reaction products form as white spots or a cloudy film as the glass surface dries. We call this phenomenon "dimming".

In varying degrees, water contact causes chemical reaction (ion exchange between cation in glass and hydrogen ion in water) which results in a silica-rich surface layer that causes an interference color at the layer. We call this phenomenon "staining".

The resistivity of glasses against these surface phenomena is expressed in terms of water durability (D_W) for dimming, and acid durability (D_A) for staining.

The classifications used in this catalog follow the revised specifications of the Japanese Optical Glass Industrial Standards (JOGIS), which differ from those in our previous catalog.

3.1 Water Durability (D_W)

Water durability is classified using the following method, into 6 groups according to percentage of weight loss:

Glass is powdered and sieved to select particle sizes of 420～590 μ. Powdered glass weighing its specific gravity is placed in a platinum net basket and soaked in 80 ml pure water (pH6.5～7.5) that is contained in a fused silica flask. The glass is then boiled for 60 minutes. The percentage of weight loss is measured.

Group	1	2	3	4	5	6
Weight Loss (%)	≦0.04	0.05～0.09	0.10～0.24	0.25～0.59	0.60～1.09	≧1.10

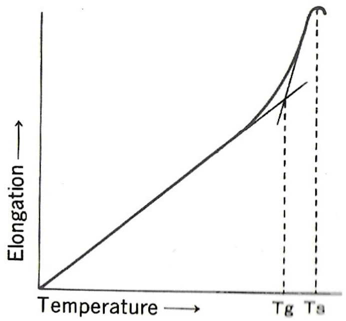
3.2 Acid Durability (D_A)

Acid durability is classified into 6 groups using a method similar to the water-durability test except that a 0.01 N nitric acid solution is used.

Group	1	2	3	4	5	6
Weight Loss (%)	≦0.19	0.20～0.34	0.35～0.64	0.65～1.19	1.20～2.19	≧2.20

4. THERMAL PROPERTIES

The thermal properties of optical glass are important for annealing and thermal processing. Three thermal properties are given. They include; transformation temperature (Tg), sag temperature (Ts), and linear coefficient of thermal expansion (α). They are determined from a thermal expansion curve obtained for a well annealed specimen, heated at a rate of 4°C/min.. A differential thermal dilatometer is used for the measurement as it maintains a uniform temperature distribution within ±1°C.



4.1 Transformation Temperature (Tg)

As illustrated in the figure, the transformation temperature is determined by the intersection point of two straight lines extended from the high and low temperature ranges of the thermal expansion curve.

4.2 Sag Temperature (Ts)

As illustrated in the figure, the sag temperature is defined as the temperature at which the thermal expansion curve starts to fall.

4.3 Linear Coefficient of Thermal Expansion (α)

The linear coefficient of thermal expansion is measured over the temperature range between 100°C and 300°C, and is expressed $10^{-7}/°C$ unit.

5. MECHANICAL PROPERTIES

Knoop hardness and abrasion factor are listed as guiding measures for lapping and polishing operations.

5.1 Knoop Hardness (Hk)

Knoop hardness is determined by loading a pyramid-shape diamond indenter with vertex angles of 172°30′ and 130° on a polished glass surface at 0.1kg for 15 seconds. It is calculated by the following formula;

$$\text{Knoop Hardness} = 14.23P/\ell^2 (kg/mm^2)$$
$$P = \text{load, (kg)}$$
$$\ell = \text{length of the longer diagonal of the indent, (mm)}$$

Group	1	2	3	4	5	6	7
Knoop Hardness kg/mm^2	≦149	150 ~249	250 ~349	350 ~449	450 ~549	550 ~649	≧ 650

5.2 Abrasion Factor (F_A)

Abrasion factor is a relative measure for lapping. A glass sample with a surface area of $9\,cm^2$ is placed 80 mm from the center of a cast iron circular plate. Then the plate is rotated horizontally at 60 r.p.m., and a 1kg lapping weight is vertically loaded on the sample. Lapping is continued for 5 minutes, with a continuous supply of a lapping compound composed of 10 g of aluminum oxide (grain size $20\,\mu$) in 20 ml of water. The weight loss of the sample is then measured and compared to that of a standard reference material (BSC 7) measured in the same way. The abrasion factor is determined by the following formula:

$$\text{Abrasion Factor} = \frac{\text{weight loss of sample glass/specific gravity}}{\text{weight loss of standard glass/specific gravity}} \times 100$$

6. OTHER

6.1 Specific Gravity

Specific gravity of glass is measured with respect to the density of pure water at 4°C.

7. QUALITY DEFINITION

7.1 Optical Tolerance

Since the listed refractive indices and Abbe-numbers are the mean of several melts, those for individual melt will somewhat differ from the mean. The tolerances are generally as follow:

Refractive index n_d : ± 0.0008

Abbe-number ν_d : $\pm 0.8\%$

Both the refractive indices at the C, d, F, and g lines and the ν_d will be furnished on melt data sheets for every annealed lot.

Upon request, we can tailor the n_d up to ± 0.0002 and the ν_d to $\pm 0.3\%$.

Your order should specify the desired tolerance.

The accuracy of "standard measurement" is ± 0.00003 for the refractive indices and ± 0.00002 for the dispersion. Upon request for "precision measurement", we can furnish a refractive index accuracy to ± 0.00002 for "i" and "t" lines, to ± 0.00001 for the rest of lines, and a dispersion accuracy to ± 0.000003.

7.2 Optical Homogeneity

The variation of refractive index in a pressed blank is generally extremely small. Within an individual lot, refractive index varies no more than ±0.0002. A tighter control of refractive index within a lot, if desired, can be proved on a special order basis.

Extremely homogenized glass of large dimension can be supplied. The glass is produced by a special melting method and is inspected with "Fizeau type" interferometer. The optical homogeneity of the glass is as follows:

Group	Variation of n_d value
H 1	$\pm 2 \times 10^{-5}$
H 2	$\pm 5 \times 10^{-6}$
H 3	$\pm 2 \times 10^{-6}$

7.3 Striae

Striae is inspected by a striae-scope that has both a point light source and an optical lens system. Our inspection follows the methods specified by JOGIS. With respect to standard glasses, striae is graded as follows:

Grade	Degree of striae
1	No visible striae
2	Light and scattered striae
3	Fine parallel striae

Upon request, inspections by MIL-G-174A methods can also be performed.

The symbol "P" appearing in the remarks column for a few glass types is an indication that they are difficult to make stria-free.

7.4 Stress Birefringence

Optical glass retains slight residual stresses even after being well annealed. Internal stresses cause birefringence, which is represented in terms of optical path difference in nm/cm.

For disc shaped products, the stress birefringence is measured at a distance 5/100 of the diameter from the outside, and, for rectangular plates, at a distance 5/100 of its width from the edge with the middle of the longer side. Stress birefringence is grouped as follows:

Grade	Stress Birefringence (nm/cm)
1	≦4 (Precision annealing)
2	5～9 (Fine annealing)
3	10～19 (Commercial annealing)
4	≧20 (Coarse annealing)

The stress birefringence in pressed blanks is normally controlled to less than 10nm/cm.

7.5 Bubbles and Inclusions

Bubbles and inclusions in our glasses, though not entirely absent, are very scarce owing to our development in melting method. The size and number of bubbles varies with the glass composition and melting conditions.

Bubbles are counted by total cross sectional area (mm^2) of bubbles for every 100 ml of glass. Inclusions such as small stones or devitrified particles are treated as bubbles and counted in the same manner. The bubbles and inclusions so counted are of size exceeding 0.05 mm in diameter. In terms of bubbles and inclusions, the quality of glass is classified as follows:

Group	1	2	3	4
Total cross sectional area per 100 ml of glass (mm^2)	≦0.11	0.12～0.24	0.25～0.49	0.50～0.99

7.6 Coloration

Coloration may appear in some glasses in spite of high purity raw materials and special melting techniques employed.

Coloration is expressed as wavelengths yielding 80% and 5% transmittance (uncorrected for surface reflection losses) in optically polished glass of 10±0.1 mm thickness. For example, a glass whose transmittance is 80% at wavelength of 340 nm, and 5% at 300 nm is designated as 34/30.

The extent of coloration varies slightly from one melt to another, and therefore the coloration listed in the catalog is the mean of several melts. Coloration between lots is controlled around ±10 nm of listed values.

For dense flint optical glasses susceptible to coloration, we are able to supply color-improved glasses made available by our special melting methods. These glass types are marked "L" in the column. If the reduced coloration is desired for these types, it is advised that you should so specify by attaching "L" after glass type designation in your order form.

Upon request, transmittance data can be furnished for delivered products.

7.7 Cadmium Free Glasses

As a result of the recent pollution concerns, we have eliminated cadmium from our CdF and CdFD glass types. These glass types are replaced both by NbF and NbFD, which contain no cadmium at all.

7.8 Thorium Free Glasses

Until recently, some of high index optical glasses, marked as "Th" in our previous catalog, contained radioactive thorium oxide. We have entirely eliminated thorium and now developed new thorium-free glasses. Designations for these glasses, however, remain unchanged.

8. FORMS OF SUPPLY

8.1 Slab

Two opposite surfaces are test polished and four sides are ground. Edges have slight bevels. The glass is fine-annealed.

8.2 Block

Two opposite surfaces are test polished, two sides ground and the remaining sides are fire-polished or cast. Edges are beveled with radii less than 15 mm. The glass is fine-annealed.

8.3 Extruded Bar

Two opposite sides, though not polished, may visually permit internal inspection, and the remaining sides are fire-polished or cast. Bevels at edges may vary depending on its dimension.

8.4 Pressed Blank

Pressed blanks in the form of lens or prism are of two types:

Hand pressed blank
Direct pressed blank

Standard tolerances are listed as follows:

Diameter (mm)	Thickness Tolerance (mm)	Diameter Tolerance (mm)
≦ 10	±0.5	±0.1
11 ~ 30	±0.4	±0.2
31 ~ 50	±0.4	±0.2
51 ~ 70	±0.4	±0.3
71 ~ 100	±0.5	±0.4
≧ 101	±0.5	±0.5

Upon request, tolerance can be held more stringently.

The dimensions of slab, block and extruded bar may differ, depending on glass type and melting methods employed. Standard dimensions are as follows:

Width	100~300 mm
Length	210~300 mm
Thickness	8~100 mm

All other sizes must be on a special order basis.

8.5 Gob

Gobs are supplied in a fire-polished form only for some glass types.

8.6 Special Shapes and Sizes

We are ready to accept special orders for sliced disc, large molded blank, window or mirror blank, and other miscellaneous shapes with various dimensions for special application.